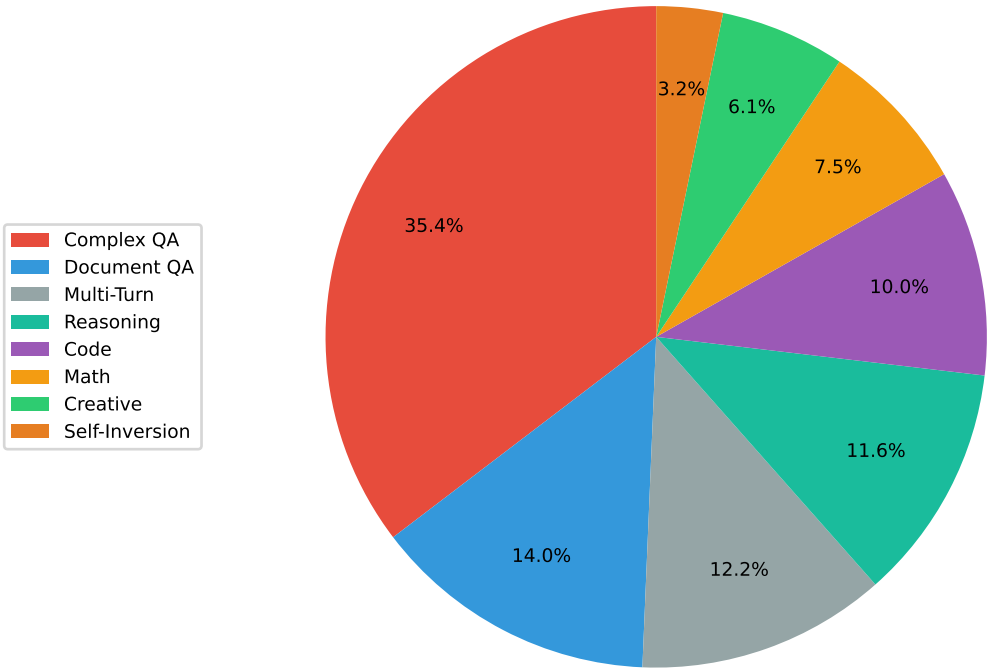
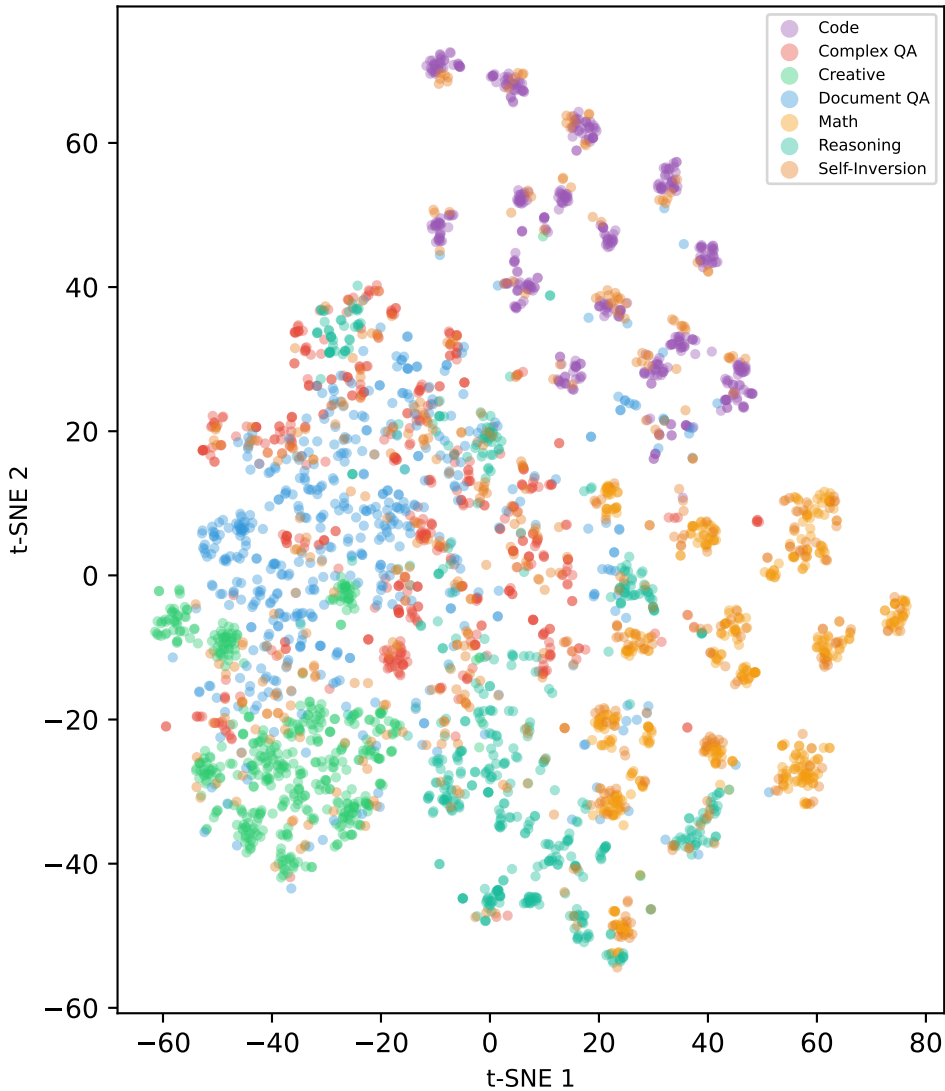


FANNO-Dev: Diversity-Verified Large-Scale Instruction Data Synthesis

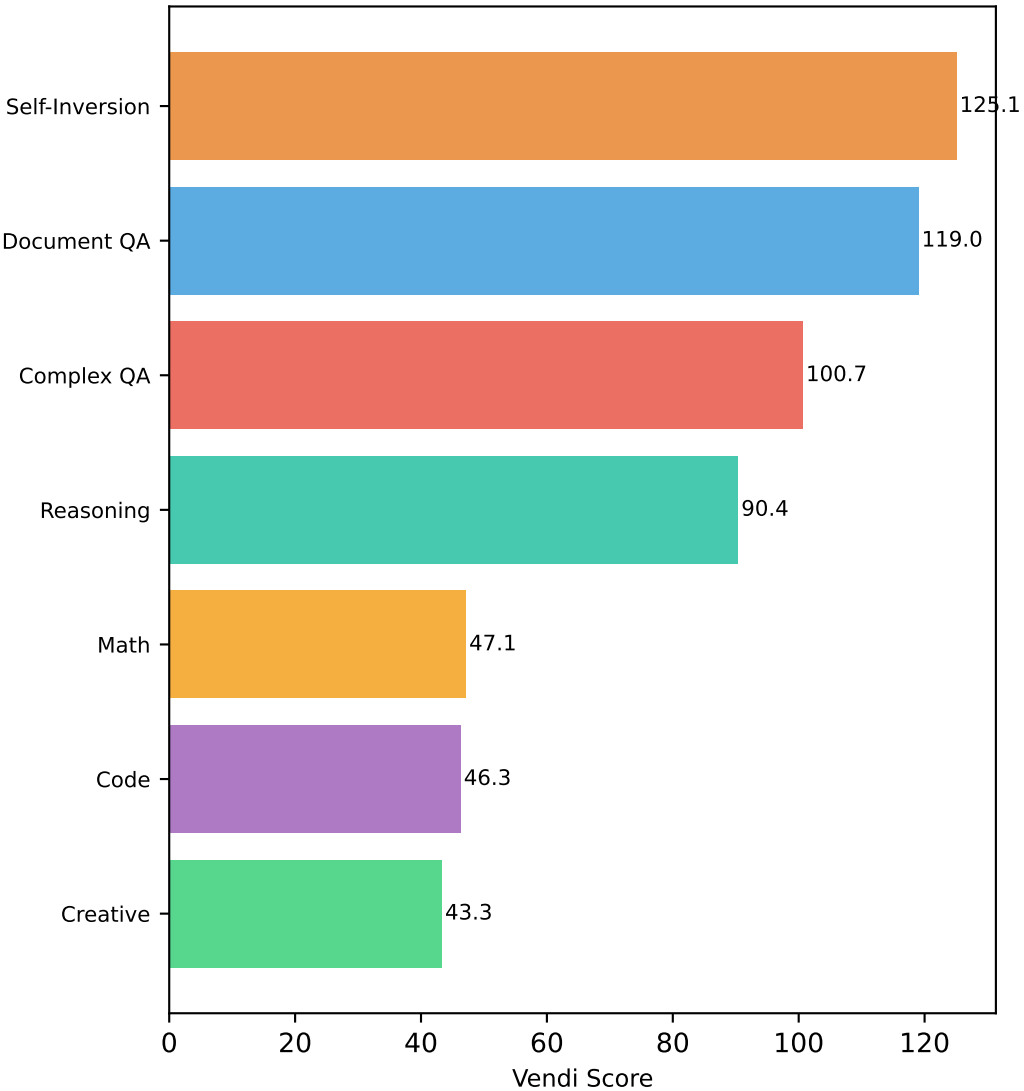
(a) Dataset Composition (153,351)



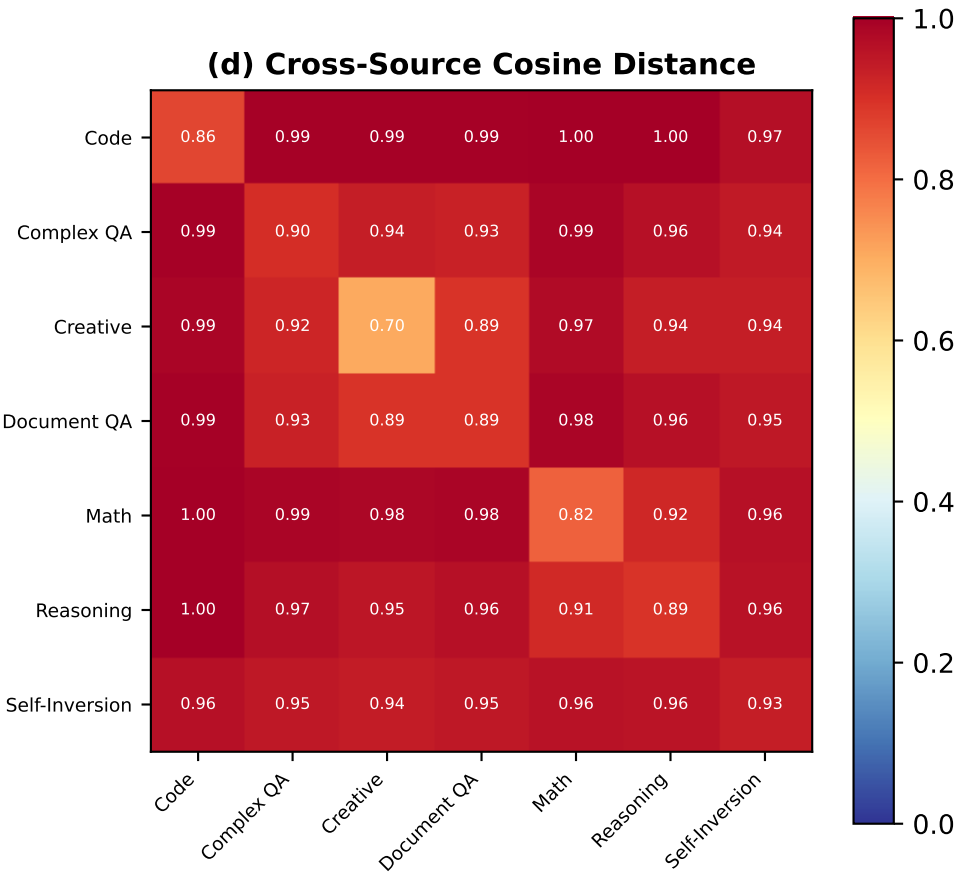
(b) Embedding Space (7 Pipelines)



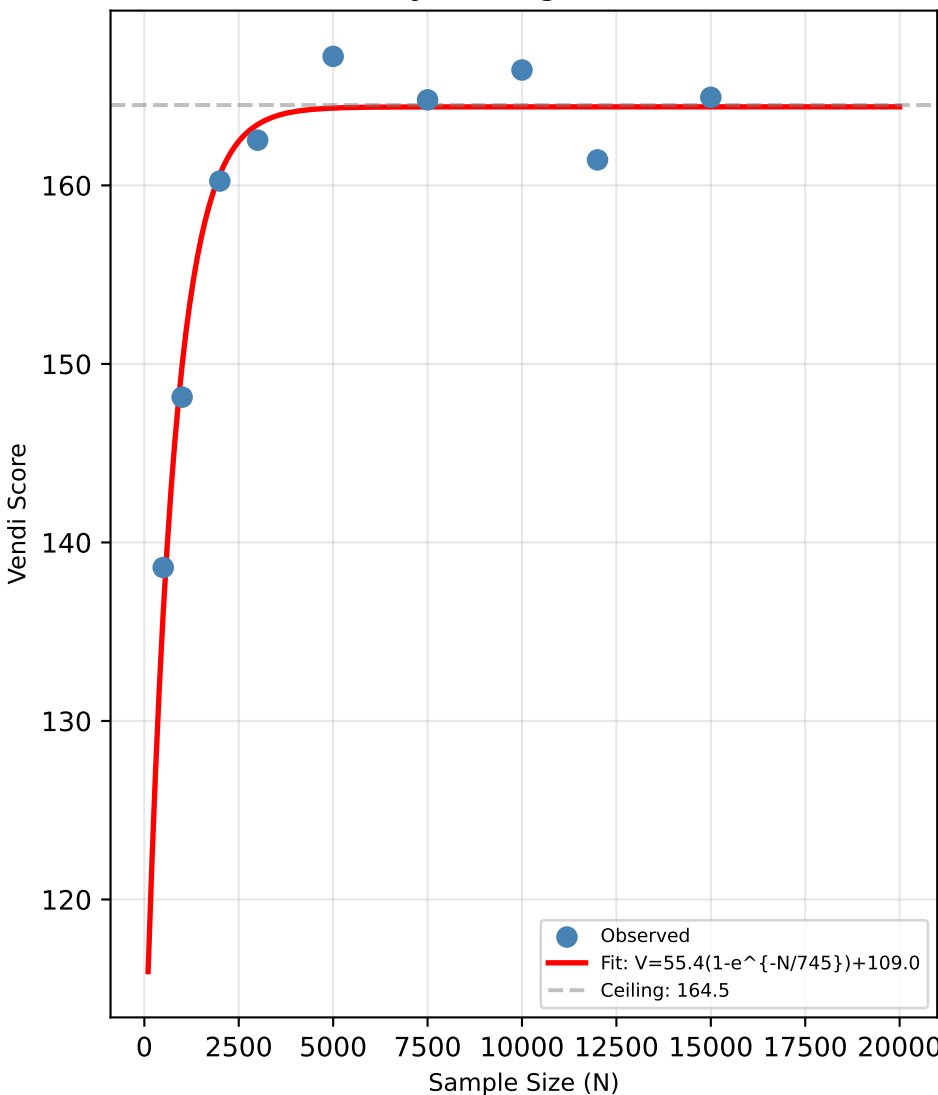
(c) Per-Pipeline Diversity



(d) Cross-Source Cosine Distance



(e) Diversity Scaling Law ( $R^2=0.996$ )



(f) Key Statistics

**153,351** cleaned samples (27.4% rejection)

**182.75** Vendi Score on 15K sample

**0.961** avg cross-source cosine distance

**2,366** unique domains (99.8% single-source)

**8** synthesis pipelines

**188** publication-quality figures

**\$2,245** total estimated cost

**0** benchmark contamination